

Finite Projection Order Defects in Residual Metric Audits: A Mathematical Starting Point for Black-Box Evaluation

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MaoField research programme

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Abstract

Modern language-model evaluation is a coupled measurement procedure rather than a transparent scalar observation: prompt distributions, benchmark axes, judge protocols, rubrics, decoding rules and aggregation weights can all enter the final score. This manuscript isolates a finite mathematical obstruction relevant to residual audits of such procedures. On a finite positive weighted two-way table $X = Q \times B$, with weighted Hilbert space $L^2(w)$, let C be the constant subspace, A the centred row-main-effect subspace and B_0 the centred column-main-effect subspace. We prove that product weights are equivalent to $A \perp B_0$, and exactly equivalent to order-independence of the two sequential stripping maps $(I - P_{B_0})(I - P_A)(I - P_C)$ and $(I - P_A)(I - P_{B_0})(I - P_C)$. Under non-product weights, there exists a pure main-effect witness whose true additive residual is zero while the wrong stripping order produces a non-zero residual-like output. An exact 2×2 rational certificate is given, with squared weighted norm $61/177408$. The MaoField part of the manuscript is programmatic only: it proposes residual metric audits as future work for studying whether observed capability, safety, reasoning, hallucination, alignment and benchmark scores are invariant model properties or coordinate-dependent artifacts of a weighted evaluation field. No empirical MaoField result is reported.

Scope and claim boundary. The established mathematical contribution of this manuscript is the finite-dimensional theorem and its exact rational witness. The MaoField sections are a research programme and set of proposed future audit objects. They do not assert a new broad ANOVA theory, a new dependent-input decomposition theory, a new theory of noncommuting projections, or an empirical finding about language models. The duplicate-risk status is MEDIUM duplicate risk. Mode B MaoField empirical status remains **insufficient_artifact**; accordingly, no MaoField empirical positive claim is made or authorized anywhere in this manuscript.

1 Introduction: residuals inside black-box evaluation

Large language model evaluation is increasingly mediated by complex measurement systems. Large benchmark suites aggregate heterogeneous tasks; holistic evaluations track multiple desiderata; open-ended generation is often judged by human raters or by other language models; and contamination, rephrasing and benchmark exposure can change what a score measures [1, 2, 6, 5, 3, 4, 7, 8, 9, 10]. The object being audited is therefore not only a model. It is a model together with a measurement field: prompts, tasks, judges, rubrics, sampling rules, weights and aggregation charts.

A central problem is residual interpretation. A residual is often read as “what remains after nuisance structure has been removed”. That reading is safe only when the residual is defined by a declared projection, and when the procedure used to compute it is stable under the relevant

weighting and chart choices. The finite theorem below shows that even in the smallest two-axis setting, sequentially removing two centred main-effect spaces can be order-dependent unless the table weights factor as a product. In that case, a pure main effect can produce a non-zero wrong-order sequential output even though its true additive residual is exactly zero.

The theorem is deliberately small. It does not explain deployed LLM systems. Its role is to provide a minimal obstruction: if residuals can become procedural artifacts in a finite weighted table, larger black-box metric systems require explicit audits of weighting, nuisance-removal order, judge protocol and aggregation chart before residual-like signals are interpreted as intrinsic model properties.

Contributions. The manuscript makes four bounded contributions.

1. It proves a finite theorem: on $Q \times B$, centred row and column main-effect spaces are orthogonal if and only if $w(q, b) = w_Q(q)w_B(b)$ cellwise.
2. It proves that two sequential main-effect stripping maps are equal if and only if the same product-weight condition holds.
3. It gives an exact 2×2 non-product rational witness in which a pure main-effect signal has zero true additive residual but non-zero wrong-order sequential output, with squared weighted norm $61/177408$.
4. It formulates a MaoField residual metric audit programme as future work: a metric residual should not be interpreted as an intrinsic model property until its invariance under weighting, nuisance-removal order, judge protocol and aggregation chart has been audited.

The last item is a proposed criterion and programme statement, not a theorem about deployed LLM systems.

2 Finite weighted setup

Let Q and B be finite non-empty sets and let $X = Q \times B$. Let $w : X \rightarrow (0, 1]$ be strictly positive and normalized:

$$\sum_{q \in Q} \sum_{b \in B} w(q, b) = 1.$$

Work in $H = \mathbb{R}^{Q \times B}$ with weighted inner product

$$\langle f, g \rangle_w = \sum_{q, b} w(q, b) f(q, b) g(q, b).$$

The row and column marginals are

$$w_Q(q) = \sum_b w(q, b), \quad w_B(b) = \sum_q w(q, b).$$

Define

$$\begin{aligned} C &= \text{span}\{\mathbf{1}\}, \\ A &= \left\{ f(q, b) = a(q) : \sum_q w_Q(q) a(q) = 0 \right\}, \\ B_0 &= \left\{ f(q, b) = c(b) : \sum_b w_B(b) c(b) = 0 \right\}. \end{aligned}$$

Let

$$N_{\text{add}} = C \oplus A \oplus B_0$$

as an algebraic direct sum. The symbol $\oplus$ is not an orthogonal direct sum unless $A \perp B_0$. Let P_C , P_A , P_{B_0} and $P_{N_{\text{add}}}$ denote the weighted orthogonal projections onto C , A , B_0 and N_{add} respectively.

The ordered stripping maps are

$$R_{Q \rightarrow B} = (I - P_{B_0})(I - P_A)(I - P_C),$$

$$R_{B \rightarrow Q} = (I - P_A)(I - P_{B_0})(I - P_C),$$

and the order-defect operator is

$$D_w = R_{Q \rightarrow B} - R_{B \rightarrow Q}.$$

Lemma 1 (Elementary orthogonality and intersection). *$C \perp A$, $C \perp B_0$, and $A \cap B_0 = \{0\}$.*

Proof. For $a \in A$, $\langle \mathbf{1}, a \rangle_w = \sum_q w_Q(q) a(q) = 0$, and the proof for B_0 is identical. If $f \in A \cap B_0$, then $f(q, b) = a(q) = c(b)$ for all (q, b) . Thus f is constant on $Q \times B$. Its weighted marginal mean is zero, so that constant is zero. $\square$

3 Finite obstruction theorem

Theorem 1 (Product weights, orthogonality and order independence). *For finite positive weights on $Q \times B$, the following are equivalent:*

- (i) $w(q, b) = w_Q(q)w_B(b)$ for every $q \in Q$ and $b \in B$;
- (ii) $A \perp B_0$ in $L^2(w)$;
- (iii) $D_w = 0$;
- (iv) $R_{Q \rightarrow B} = R_{B \rightarrow Q}$.

When these conditions hold,

$$R_{Q \rightarrow B} = R_{B \rightarrow Q} = I - P_{N_{\text{add}}}.$$

If they fail, then there exists a pure main-effect witness $K \in A$ or $K \in B_0$ such that

$$(I - P_{N_{\text{add}}})K = 0,$$

one ordered stripping output is zero, and the other ordered stripping output is non-zero.

Proof. First suppose $w(q, b) = w_Q(q)w_B(b)$. For $a \in A$ and $c \in B_0$,

$$\langle a, c \rangle_w = \sum_{q, b} w_Q(q)w_B(b)a(q)c(b) = \left(\sum_q w_Q(q)a(q) \right) \left(\sum_b w_B(b)c(b) \right) = 0.$$

Thus $A \perp B_0$.

Conversely assume $A \perp B_0$. For fixed q_0, b_0 , define centred indicators

$$a_{q_0}(q) = \mathbf{1}_{q=q_0} - w_Q(q_0), \quad c_{b_0}(b) = \mathbf{1}_{b=b_0} - w_B(b_0).$$

They lie in A and B_0 . Their inner product is

$$\langle a_{q_0}, c_{b_0} \rangle_w = w(q_0, b_0) - w_Q(q_0)w_B(b_0).$$

Orthogonality therefore gives the product identity for every cell. This proves (i) $\Leftrightarrow$ (ii).

Since $P_A \mathbf{1} = P_{B_0} \mathbf{1} = 0$,

$$D_w = (I - P_{B_0})(I - P_A)(I - P_C) - (I - P_A)(I - P_{B_0})(I - P_C) = P_{B_0}P_A - P_AP_{B_0}.$$

If $A \perp B_0$, then $P_AP_{B_0} = P_{B_0}P_A = 0$, so $D_w = 0$.

Conversely suppose $D_w = 0$. For any $b \in B_0$,

$$0 = D_w b = P_{B_0}P_A b - P_A b.$$

Thus $P_A b \in A$ and $P_A b = P_{B_0}P_A b \in B_0$. By the lemma, $P_A b = 0$. Because P_A is the orthogonal projection onto A , every $b \in B_0$ is orthogonal to A . Hence $A \perp B_0$. This proves (ii) $\Leftrightarrow$ (iii), and (iii) $\Leftrightarrow$ (iv) follows from the definition of D_w .

When $A \perp B_0$, the subspaces C, A, B_0 are mutually orthogonal and $P_{N_{\text{add}}} = P_C + P_A + P_{B_0}$. The ordered products therefore reduce to $I - P_{N_{\text{add}}}$.

It remains to prove the existential witness statement under non-product weights. If the weights are non-product, then A and B_0 are not orthogonal. Therefore there exists $K \in B_0$ with $P_A K \neq 0$; otherwise all of B_0 would be orthogonal to A . Since $K \in B_0$, $P_C K = 0$ and $P_{B_0} K = K$. Hence

$$R_{B \rightarrow Q} K = (I - P_A)(I - P_{B_0})(I - P_C)K = 0.$$

On the other hand,

$$R_{Q \rightarrow B} K = (I - P_{B_0})(I - P_A)K = -(I - P_{B_0})P_A K.$$

If this vector were zero, then $P_A K = P_{B_0}P_A K$ would lie in $A \cap B_0 = \{0\}$, contradicting $P_A K \neq 0$. Thus $R_{Q \rightarrow B} K \neq 0$. Finally $K \in B_0 \subset N_{\text{add}}$, so $(I - P_{N_{\text{add}}})K = 0$. The non-zero wrong-order output is therefore not a true additive residual. $\square$

4 Exact rational certificate

Let

$$w = \frac{1}{11} \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix},$$

with cells ordered as $((q_1, b_1), (q_1, b_2), (q_2, b_1), (q_2, b_2))$. The marginals are

$$w_Q = \left(\frac{3}{11}, \frac{8}{11} \right), \quad w_B = \left(\frac{4}{11}, \frac{7}{11} \right).$$

The weight is not product form because

$$w(q_1, b_1) = \frac{1}{11} \neq \frac{12}{121} = w_Q(q_1)w_B(b_1).$$

Consider the pure column-main-effect witness

$$K = \left(\frac{7}{11}, -\frac{4}{11}, \frac{7}{11}, -\frac{4}{11} \right) \in B_0.$$

It belongs to N_{add} , so

$$(I - P_{N_{\text{add}}})K = 0.$$

The column-first order removes it immediately:

$$R_{B \rightarrow Q} K = 0.$$

The row-first order gives

$$R_{Q \rightarrow B}K = \left(\frac{1}{32}, \frac{5}{168}, -\frac{1}{96}, -\frac{1}{84} \right).$$

Consequently

$$\|R_{Q \rightarrow B}K\|_w^2 = \frac{1}{11} \left(\frac{1}{32} \right)^2 + \frac{2}{11} \left(\frac{5}{168} \right)^2 + \frac{3}{11} \left(\frac{1}{96} \right)^2 + \frac{5}{11} \left(\frac{1}{84} \right)^2 = \frac{61}{177408}.$$

This is an exact rational certificate, not a floating-point experiment.

5 What the theorem does and does not imply

The theorem proves a finite obstruction to a common residual interpretation. It separates two quantities that coincide under product weights but need not coincide under non-product weights:

true additive residual: $(I - P_{N_{\text{add}}})K$,

sequential stripping output: $R_{Q \rightarrow B}K$ or $R_{B \rightarrow Q}K$.

The exact witness has zero true additive residual and non-zero wrong-order sequential output. The output is therefore an artifact of the stripping procedure, not evidence of a true interaction term.

The theorem does not prove that any deployed language-model metric currently contains such an artifact. It does not identify a hallucination mechanism, a reasoning circuit, an alignment mechanism, a benchmark contamination mechanism or a judge-bias mechanism. It proves that residual-like quantities can be order-dependent in a finite weighted nuisance-removal chart. That is enough to motivate explicit residual audits, but not enough to report an empirical MaoField finding.

6 MaoField programme: metric complexes as audit objects

This section is programmatic. It proposes audit objects for later work; it does not report observations from a MaoField empirical panel.

6.1 Metrics as fields rather than transparent scores

A language-model score can be represented schematically as

$$\mathcal{M}(\text{model}, \text{prompt distribution}, \text{task}, \text{judge}, \text{rubric}, \text{decoding}, \text{weighting}, \text{aggregation}).$$

The MaoField programme treats this composite as an audit object. The relevant black box is not only the model. The metric apparatus can also be black-boxed: a change in score may reflect model behaviour, judge bias, rubric drift, sampling design, benchmark contamination, temporal drift, aggregation weights or their couplings.

A residual metric audit asks whether a reported residual-like signal is invariant under declared changes of weighting, nuisance-removal order, judge protocol and aggregation chart. If not, the signal should be treated as a candidate measurement artifact until further evidence separates model behaviour from metric design.

6.2 Candidate finite audit charts

A first finite audit can use two axes:

$$Q = \{\text{query, prompt or task classes}\}, \quad B = \{\text{judge, rubric, benchmark or context classes}\}.$$

Let $w(q, b)$ be the designed or empirical cell weight, and let $K(q, b)$ be a scalar diagnostic such as score, error rate, preference margin, refusal rate, hallucination indicator, calibration gap or safety violation rate. The finite theorem then asks whether residuals are computed by a true projection relative to a declared nuisance model, or by an order-dependent sequential procedure.

This yields concrete future audits:

1. **Weight-form audit:** measure how far $w(q, b)$ is from $w_Q(q)w_B(b)$ before interpreting residuals.
2. **Order-defect audit:** compute $R_{Q \rightarrow B}K - R_{B \rightarrow Q}K$ as procedural sensitivity, not as a capability score.
3. **Projection audit:** compare sequential outputs against the true projection residual $(I - P_{N_{\text{add}}})K$.
4. **Judge-axis audit:** treat human judges, model judges, rubrics and answer-order protocols as axes, not as neutral windows.
5. **Contamination-axis audit:** represent benchmark exposure, paraphrase exposure and fresh-item conditions as conditioning or weighting axes.
6. **Aggregation audit:** test whether a leaderboard residual survives changes from micro-average to macro-average, stratified weights or temporally updated weights.

7 Future mathematical objects

The theorem suggests several mathematical objects for future work.

1. **Multi-axis finite audit cubes.** Replace $Q \times B$ by $X_1 \times \cdots \times X_m$ and study when nuisance-removal paths produce invariant residuals.
2. **Near-product perturbation bounds.** Bound $\|D_w\|$ by a dependence measure such as $\|w - w_Q \otimes w_B\|$ under finite positivity conditions.
3. **Commutator norm instability indices.** Use $\|P_{B_0}P_A - P_AP_{B_0}\|$ as a chart-instability diagnostic.
4. **Residual chart equivalence classes.** Identify which nuisance-removal charts yield the same projection residuals.
5. **Weight-sensitive rank stability.** Study when model rankings remain stable under changes of cell weighting.
6. **Judge-order defects.** Treat answer order, judge identity, verbosity, self-enhancement and rubric placement as axes capable of producing order defects.
7. **Benchmark-contamination coupling certificates.** Build exact finite examples separating model change from contamination-weight coupling.
8. **Exact certificate libraries.** Collect rational counterexamples so that audit failures can be reproduced without relying on JSON floats.

These are future mathematical problems. The present manuscript proves only the two-axis finite theorem above.

8 Related work and duplicate risk

This manuscript is adjacent to several established literatures. Dependent-input functional decompositions and sensitivity analysis study attribution when inputs are statistically dependent [11, 12, 13, 14, 15, 16, 17]. Classical two-subspace and projection-product theory studies products, commutators and geometry of projections in greater generality [18, 19, 20]. Language-model evaluation research has developed broad benchmark suites, holistic evaluation frameworks, model-based judging, truthfulness and hallucination diagnostics, trustworthiness audits and contamination studies [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

The contribution here is narrower than all of these areas: a finite weighted two-way projection-order artifact with an exact rational certificate, plus a programme for applying residual-audit discipline to metric complexes. It does not claim a broad theory of ANOVA, dependent-input decomposition, sensitivity analysis, projection products or language-model evaluation. The duplicate-risk status is MEDIUM duplicate risk. A final bibliography pass should verify all metadata and author spellings against authoritative external records before any public release path is considered.

9 Limitations and reproducibility

The limitations are structural.

1. The theorem is finite-dimensional and two-axis. It does not cover arbitrary multi-axis evaluation systems.
2. The theorem concerns orthogonal projections in $L^2(w)$, not causal effects, semantic truth, mechanistic circuits or human preference.
3. The exact witness is a certificate of possibility, not evidence about an actual language model.
4. The MaoField programme is future work. Mode B MaoField empirical status remains `insufficient_artifact`. No MaoField empirical positive claim is made.
5. The floating-point harness is deterministic regression support only; the mathematical claims are carried by the analytic proof and exact rational certificate, not by JSON floats.

All mathematical statements in the theorem are proved analytically in the text. The 2×2 witness is rational and can be checked by exact arithmetic from the displayed weight matrix, witness vector and projection definitions. The floating-point harness is deterministic regression support only; the mathematical claims are carried by the analytic proof and exact rational certificate, not by JSON floats.

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